

Predicting Earth-Like Similarity in Exoplanets using Machine Learning Experimental Results

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Experiment #1: Baseline Ranking Comparison Study

The purpose of the Experiment #1: Baseline Ranking Comparison Study is to show that our proposed method weighted Earth-like similarity ranking provides more meaningful results than simpler ranking approaches. These simpler baseline K-Nearest Neighbor (KNN) models will be unweighted where all feature weights = 1. This will allow us to determine if the same strong Earth-like candidates remain near the top even when simpler methods are used. If the top-ranked candidates remain stable, that supports the robustness of the method. If some rankings change, we can analyze which features or weighting choices caused the difference.

Therefore, our weighted proposed method KNN models #1 and #2 will be compared to unweighted baseline KNN models #3 and #4 using the Manhattan and Euclidean distance metrics respectively. Then the same weighted proposed method KNN models #1 and #2 will be compared to unweighted baseline KNN models #5 and #6 which is filtered for exoplanets whose habitable zone flag = 1. This will also use the Manhattan and Euclidean distance metrics respectively. The comparison of the KNN models #5 and #6 will allow us to isolate the effect of applying the habitable zone filter alone, without introducing feature weighting.

Experiment #1: Baseline Ranking Comparison Study						
	Model #1	Model #2	Model #3	Model #4	Model #5	Model #6
Rank	Weighted Proposed Method Manhattan	Weighted Proposed Method Euclidean	Unweighted Baseline Manhattan	Unweighted Baseline Euclidean	Unweighted Habitable Zone Only Baseline Manhattan	Unweighted Habitable Zone Only Baseline Euclidean
1	Kepler-452 b	Kepler-452 b	Kepler-452 b	Kepler-1185 b	Kepler-438 b	Kepler-438 b
2	Kepler-1638 b	Kepler-1638 b	Kepler-1185 b	Kepler-1253 b	Kepler-296 e	Kepler-442 b
3	Kepler-51 e	Kepler-51 e	Kepler-132 e	Kepler-132 e	Kepler-442 b	Kepler-395 c
4	Kepler-1606 b	Kepler-1606 b	Kepler-1126 c	Kepler-384 c	Kepler-395 c	Kepler-296 e
5	Kepler-1701 b	Kepler-1701 b	Kepler-384 c	Kepler-130 d	Kepler-138 e	Kepler-138 e
6	Kepler-22 b	Kepler-1126 c	Kepler-1638 b	Kepler-1440 b	Kepler-452 b	Kepler-452 b
7	Kepler-1126 c	Kepler-69 c	Kepler-1253 b	Kepler-1498 b	Kepler-1410 b	Kepler-1410 b
8	Kepler-1549 b	Kepler-1185 b	Kepler-1965 b	Kepler-1441 b	Kepler-1544 b	Kepler-1544 b
9	Kepler-1690 b	Kepler-442 b	Kepler-1440 b	Kepler-1965 b	Kepler-1701 b	Kepler-1701 b
10	Kepler-69 c	Kepler-311 d	Kepler-1605 b	Kepler-1620 b	Kepler-1606 b	Kepler-1450 b
Top 5 overlap between weighted proposed method & unweighted baseline			1/5	0/5	0/5	0/5
Top 10 overlap between weighted proposed method & unweighted baseline			3/10	1/10	3/10	3/10

Table 1: Displays Experiment #1 Baseline Study Models which compares the Weighted Proposed Method KNN Models #1 and #2 with Unweighted Baseline KNN Models #3, #4, #5, and #6

The results of the Experiment #1: Baseline Ranking Comparison Study referenced in Table 1. When comparing the Weighted Proposed Method Manhattan KNN Model #1 with Unweighted Baseline Manhattan KNN Model #3 to determine the effects of feature weighting, only a small portion of top Earth-Like exoplanets remained the same when weights were removed. Only Kepler-452 b remained the most Earth-Like when weights were removed however no one top 5 exoplanet remained consistent. Overall, 3 exoplanets remained in the top 10 list. These results indicate that the features weights proposed are not redundant but rather strongly influence final rankings. These results also indicate that unweighted Manhattan distance may overvalue some parameters or define all parameters as equally relevant for Earth-Like similarity and potential habitability. Our proposed method, however, better prioritizes Earth-relevant parameters.

A similar trend can be observed when comparing the Weighted Proposed Method Euclidean KNN Model #2 with Unweighted Baseline Euclidean KNN Model #4. When features weights were removed, none of the top 5 exoplanets remained and only 1 exoplanet remained in the top 10 list, showing more significant changes compared to the Manhattan distance metric results. This shows that the Euclidean distance metric is highly sensitive to scaling and feature weighting. Therefore, the proposed weighted Euclidean distance may be able to correct the distortions caused by the unweighted Euclidean distance, resulting in a more stabilized ranking.

When comparing the Weighted Proposed Method Manhattan KNN Model #1 with the Unweighted Habitable Zone Only Baseline Manhattan KNN Model #5, none of the top 5 exoplanets remained consistent however only 3 exoplanets remained the same in both top 10 lists. The same results appear when comparing the Weighted Proposed Method Euclidean KNN Model #2 with the Unweighted Habitable Zone Only Baseline Euclidean KNN Model #6. These observations show that simple habitable zone filtering does not produce the same results as our proposed weighted rankings. These results indicate that only filtering for exoplanets who are within the habitable zone of their host stars alone is not sufficient in determining Earth-Like similarity. This shows that other relevant Earth-Like parameters such as planetary radius and temperature, orbital distance, orbital period, and stellar properties are equally important. Therefore, our proposed weighted features capture multi-factor Earth similarity rather than just location in orbit.

An important observation to note is that the exoplanets Kepler-452 b appears as the most robust candidate since it remained the most consistent across all ranking methods except for Unweighted Baseline Euclidean KNN Model #4, where its ranking dropped to 15th place. These results verify that Kepler-452 b is the closest match to Earth with a potential for habitability. In addition to Kepler-452 b, the exoplanets Kepler-1638 b, Kepler-1126 c, Kepler-1606 b, Kepler-1701 b, Kepler-1185 b, and Kepler-442 b appeared most frequently across various ranking methods showing that these exoplanets are relatively robust candidates.

Experiment #2: Ablation Study

The purpose of the Experiment #2: Ablation Study is to identify which feature groups are most important for the Earth-like ranking. The same unweighted ranking method will be run for both the Manhattan and Euclidean distance metrics but each time, a different group of features will be removed to observe how the top Earth-Like exoplanets change and to determine which features are the most relevant for Earth-Like similarity ranking.

The Weighted Proposed Method Manhattan KNN Model #7 and Weighted Proposed Method Euclidean KNN Model #8 represent the Full Model group where all features are selected with proposed weights applied. All other models for the Experiment #2: Ablation Study will be compared to KNN Models #7 and #8 based on the Manhattan and Euclidean distance metric respectively.

The No Spectral Type group is represented by the Unweighted No Spectral Type Manhattan KNN Model #9 and Unweighted No Spectral Type Euclidean KNN Model #10 where only the one hot encoded spectral type columns are removed. The No Stellar Features (Partial) group is represented by the Unweighted No Stellar Features (Partial) Manhattan KNN Model #11 and Unweighted No Stellar Features (Partial) Euclidean KNN Model #12 where all stellar features are removed but one hot encoded spectral type features are still included. The stellar features removed include stellar mass, stellar radius, stellar effective temperature, stellar luminosity, and stellar surface gravity. Likewise, the No Stellar Features (Full) group is represented by the Unweighted No Stellar Features (Full) Manhattan KNN Model #11.5 and Unweighted No Stellar Features (Full) Euclidean KNN Model #12.5 where all stellar and one hot encoded spectral type features are removed.

The No Planetary Features group is represented by the Unweighted No Planetary Features Manhattan KNN Model #13 and Unweighted No Planetary Features Euclidean KNN Model #14. The planetary features removed include planet radius, planet mass, planet equilibrium temperature, and planetary orbital velocity. The No Orbital Features group is represented by the Unweighted No Orbital Features Manhattan KNN Model #15 and Unweighted No Orbital Features Euclidean KNN Model #16. The orbital features removed include orbital period and orbital distance (orbital semi major axis). The No Habitable Zone group is represented by the Unweighted No Habitable Zone Manhattan KNN Model #17 and Unweighted No Habitable Zone Euclidean KNN Model #17 where the habitable zone binary column is removed. Lastly, the No Temperature group is represented by the Unweighted No Temperature Manhattan KNN Model #19 and Unweighted No Temperature Euclidean KNN Model #20 where the planetary equilibrium temperature and stellar effective temperature features were removed.

Experiment #2: Ablation Study								
	Full Model		No Spectral Type		No Stellar Features (Partial)		No Stellar Features (Full)	
	Model #7	Model #8	Model #9	Model #10	Model #11	Model #12	Model #11.5	Model #12.5
Rank	Weighted Proposed Method Manhattan	Weighted Proposed Method Euclidean	Unweighted No Spectral Type Manhattan	Unweighted No Spectral Type Euclidean	Unweighted No Stellar Features Manhattan (Partial)	Unweighted No Stellar Features Euclidean (Partial)	Unweighted No Stellar Features Manhattan (Full)	Unweighted No Stellar Features Euclidean (Full)
1	Kepler-452 b	Kepler-452 b	Kepler-452 b	Kepler-1185 b	Kepler-452 b	Kepler-1185 b	Kepler-442 b	Kepler-442 b
2	Kepler-1638 b	Kepler-1638 b	Kepler-1185 b	Kepler-1253 b	Kepler-1185 b	Kepler-132 e	Kepler-438 b	Kepler-186 f
3	Kepler-51 e	Kepler-51 e	Kepler-442 b	Kepler-132 e	Kepler-1605 b	Kepler-442 b	Kepler-186 f	Kepler-438 b
4	Kepler-1606 b	Kepler-1606 b	Kepler-132 e	Kepler-1620 b	Kepler-442 b	Kepler-186 f	Kepler-452 b	Kepler-138 e
5	Kepler-1701 b	Kepler-1701 b	Kepler-1126 c	Kepler-384 c	Kepler-438 b	Kepler-1253 b	Kepler-138 e	Kepler-1185 b
6	Kepler-22 b	Kepler-1126 c	Kepler-438 b	Kepler-130 d	Kepler-132 e	Kepler-1498 b	Kepler-395 c	Kepler-395 c
7	Kepler-1126 c	Kepler-69 c	Kepler-1620 b	Kepler-1440 b	Kepler-1126 c	Kepler-438 b	Teegarden's Star d	Kepler-1229 b
8	Kepler-1549 b	Kepler-1185 b	Kepler-384 c	Kepler-1498 b	Kepler-384 c	Kepler-384 c	Teegarden's Star c	Teegarden's Star d
9	Kepler-1690 b	Kepler-442 b	Kepler-1638 b	Kepler-442 b	Kepler-1965 b	Kepler-1965 b	TOI-700 e	TOI-700 d
10	Kepler-69 c	Kepler-311 d	Kepler-1253 b	Kepler-1459 b	Kepler-186 f	Kepler-138 e	TOI-700 d	GJ 1002 c
Top 5 overlap between weighted proposed method & unweighted ablation			1/5	0/5	1/5	0/5	0/5	0/5
Top 10 overlap between weighted proposed method & unweighted ablation			3/10	2/10	2/10	2/10	1/10	2/10

Table 2: Displays Experiment #2 Ablation Study Models which compares the Weighted Proposed Method KNN Models #7 and #8 with the Unweighted Ablation KNN Models #9, #10, #11, #12, #11.5, and #12.5

The results of the Experiment #1: Ablation Study referenced in Table 2 and Table 3. The results in Table 2 show that when comparing the Full Model to the No Spectral Type group, there appears to be significant changes in both the Manhattan and Euclidean distances when the spectral type columns are removed. This implies that the spectral type features contribute significantly to Earth-Like ranking stability. When comparing the Full Model to the No Stellar Feature (Partial) group, the results show that even with spectral type features remaining, removing stellar mass, radius, temperature, luminosity, and gravity cause major ranking disruption. This suggests continuous stellar variables are important features for determining Earth-Like similarity. When comparing the Full Model to the No Stellar Feature (Full) group, the results show that removing all stellar and spectral type information causes the strongest disruptions in similarity rankings. This implies that the host star properties are among the most critical determinants of Earth-like similarity. This can be expected since the stellar features paired with the spectral type, describing star type and luminosity class, provide strong insight into the stellar lifespan, solar flare activity, radiation environment, long-term planetary climate stability, and habitability zone location. These are all critical parameters for Earth-Like surface composition and potential habitability.

Experiment #2: Ablation Study										
	Full Model		No Planetary Features		No Orbital Features		No Habitable Zone		No Temperature	
	Model #7	Model #8	Model #13	Model #14	Model #15	Model #16	Model #17	Model #18	Model #19	Model #20
Rank	Weighted Proposed Method Manhattan	Weighted Proposed Method Euclidean	Unweighted No Planetary Features Manhattan	Unweighted No Planetary Features Euclidean	Unweighted No Orbital Features Manhattan	Unweighted No Orbital Features Euclidean	Unweighted No Habitable Zone Manhattan	Unweighted No Habitable Zone Euclidean	Unweighted No Temperature Manhattan	Unweighted No Temperature Euclidean
1	Kepler-452 b	Kepler-452 b	HD 43197 b	Kepler-1625 b	Kepler-1185 b	Kepler-1253 b	Kepler-1185 b	Kepler-1185 b	Kepler-452 b	Kepler-1570 b
2	Kepler-1638 b	Kepler-1638 b	Kepler-1638 b	Kepler-1638 b	Kepler-452 b	Kepler-1185 b	Kepler-452 b	Kepler-1253 b	Kepler-1185 b	Kepler-1253 b
3	Kepler-51 e	Kepler-51 e	Kepler-452 b	Kepler-784 b	Kepler-384 c	Kepler-384 c	Kepler-132 e	Kepler-132 e	Kepler-1605 b	Kepler-1185 b
4	Kepler-1606 b	Kepler-1606 b	Kepler-289 e	Kepler-1570 b	Kepler-132 e	Kepler-132 e	Kepler-1126 c	Kepler-384 c	Kepler-384 c	Kepler-384 c
5	Kepler-1701 b	Kepler-1701 b	Kepler-1625 b	Kepler-1929 b	Kepler-1965 b	Kepler-1440 b	Kepler-384 c	Kepler-130 d	Kepler-1440 b	Kepler-1440 b
6	Kepler-22 b	Kepler-1126 c	PH2 b	Kepler-525 b	Kepler-1440 b	Kepler-1498 b	Kepler-1253 b	Kepler-1440 b	Kepler-1570 b	Kepler-132 e
7	Kepler-1126 c	Kepler-69 c	Kepler-1634 b	Kepler-638 b	Kepler-1253 b	Kepler-130 d	Kepler-1965 b	Kepler-1498 b	Kepler-132 e	Kepler-1605 b
8	Kepler-1549 b	Kepler-1185 b	Kepler-1636 b	HD 43197 b	Kepler-1505 b	Kepler-1441 b	Kepler-1440 b	Kepler-1441 b	Kepler-1253 b	Kepler-130 d
9	Kepler-1690 b	Kepler-442 b	HD 142415 b	HD 44219 b	Kepler-1498 b	Kepler-1965 b	Kepler-1605 b	Kepler-1965 b	Kepler-1505 b	Kepler-1441 b
10	Kepler-69 c	Kepler-311 d	Kepler-1519 b	Kepler-806 b	Kepler-1441 b	Kepler-1505 b	Kepler-1505 b	Kepler-1620 b	Kepler-1965 b	Kepler-1498 b
Top 5 overlap between weighted proposed method & unweighted ablation			2/5	1/5	1/5	0/5	1/5	0/5	1/5	0/5
Top 10 overlap between weighted proposed method & unweighted ablation			2/10	1/10	1/10	1/10	2/10	1/10	1/10	1/10

Table 3: Displays Experiment #2 Ablation Study Models which compares the Weighted Proposed Method KNN Models #7 and #8 with the Unweighted Ablation KNN Models #13, #14, #15, #16, #17, #18, #19, and #20.

The results in Table 3 show that when comparing the Full Model to the No Planetary Features group, there appears to be strong rankings shifts when planetary features are removed. However, its effects are surprisingly less significant than when all stellar and spectral type features are removed. This implies that both planet and host star features are jointly critical for determining Earth-Like similarity rather than planetary features alone.

Comparing the Full Model to the No Orbital Features group and the No Temperature group respectfully reveals that removing orbital or temperature features had equally the strongest effects on ranking stability due to the low overlap values for the top 5 and top 10 lists. This can be expected since orbital properties along with planetary and stellar temperatures are critical for determining habitability conditions and planet atmospheric composition. Additionally, given only about 1.2% of all confirmed exoplanets are found to have an orbital period similar to Earth's between 300-400 days, orbital period is therefore one of the most rare and distinctive features needed for Earth-Like similarity assessment. This effect may also be influenced by detection bias, as

longer period exoplanets remain underrepresented in current catalogs. Lastly, comparing the Full Model to the No Habitable Zone group revealed that removing this binary feature causes notable changes to the top 5 and top 10 results. This shows that habitable zone contributes meaningful ranking power but is not sufficient alone based on Experiment #1 results.

Both Table 2 and Table 3 show that the top 5 overlap and top 10 overlap remain very low (between 0-3 overlapping planets) across all ranking methods when compared to full weighted model. This indicates that Earth-Like similarity ranking depends on multiple interacting feature groups rather than a single feature or feature group. This therefore supports the validity of the proposed method for a weighted full model.

The features which had the highest importance were orbital features, temperature features, and host star features while planetary features and habitable zone had strong importance. This indicates that host star environment and thermal conditions are central to Earth similarity assessment. Kepler-452 b once again remained the most robust in rankings followed by Kepler-1638 b, Kepler-1126 c, Kepler-1185 b, and Kepler-442 b which supports the results from Experiment #1.

Across nearly all ablations, Manhattan distance retained more overlap and thus, exhibited greater ranking stability than Euclidean distance. This can be expected due to the sensitivity of the Euclidean distance metric. This suggests improved robustness for combined exoplanet feature similarity.

Experiment #3: Weight Sensitivity Analysis

The purpose of this experiment is to test whether the top Earth-like candidates remain stable when the feature weights were slightly altered by $\pm 20\%$. The Original Weights Model is represented by the Weighted Proposed Method Manhattan KNN Model #7 and Weighted Proposed Method Euclidean KNN Model #8. Then several modified weight settings were created and rerun to analyze the top 10 rankings.

Weighted +20% Planetary Features Method Manhattan KNN Model #21 and Weighted +20% Planetary Features Method Euclidean KNN Model #22 used a weight setting which increased the planetary feature weights by 20%. No weight was applied to other features. The planetary features included planet radius, planet mass, planet equilibrium temperature, and planetary orbital velocity. Similarity, Weighted -20% Planetary Features Method Manhattan KNN Model #23 and Weighted -20% Planetary Features Method Euclidean KNN Model #24 used a weight setting which decreased the planetary feature weights by 20% while all other features had no weights applied.

Weighted +20% Full Stellar Features Method Manhattan KNN Model #25 and Weighted +20% Full Stellar Features Method Euclidean KNN Model #26 used a weight setting which increased all the stellar feature weights by 20%. No weight was applied to other features. The full stellar features included stellar mass, stellar radius, stellar effective temperature, stellar luminosity, stellar surface gravity, and one hot encoded spectral type columns. Similarly, Weighted -20% Full Stellar Features Method Manhattan KNN Model #27 and Weighted -20% Full Stellar Features Method Euclidean KNN Model #28 used a weight setting which decreased the planetary feature weights by 20% while all other features had no weights applied.

Weighted +20% Partial Stellar Features Method Manhattan KNN Model #29 and Weighted +20% Partial Stellar Features Method Euclidean KNN Model #30 used a similar weight setting as KNN Models #25 and #26 respectively. Additionally, Weighted -20% Partial Stellar Features Method Manhattan KNN Model #31 and Weighted -20% Partial Stellar Features Method Euclidean KNN Model #32 used a similar weight setting as KNN Models #27 and #28 respectively. However, the one hot spectral type columns were not applied with weights for KNN Models #29, #30, #31, and #32.

Weighted +20% Spectral Type Features Method Manhattan KNN Model #31 and Weighted +20% Spectral Type Features Method Euclidean KNN Model #32 applies an increased 20% weight for all one-hot encoded spectral-type features only. Similarly, Weighted -20% Spectral Type Features Method Manhattan KNN Model #33 and Weighted -20% Spectral Type Features Method Euclidean KNN Model #34 applies a decreased 20% weight for all one-hot encoded spectral-type features only.

Lastly, Weighted +20% Orbital and Habitable Zone Features Method Manhattan KNN Model #31 and Weighted +20% Orbital and Habitable Zone Features Method Euclidean KNN Model #32 applies an increased 20% weight for the orbital period, orbital semi-major axis, and habitable-zone flag features only. Similarly, Weighted -20% Orbital and Habitable Zone Features Method Manhattan KNN Model #33 and Weighted -20% Orbital and Habitable Zone Features Method Euclidean KNN Model #34 applies an decreased 20% weight for all orbital and habitable zone features.

Experiment #3: Weight Sensitivity Analysis						
	Original Weights Model		Planetary Features +20% Weight		Planetary Features -20% Weight	
	Model #7	Model #8	Model #21	Model #22	Model #23	Model #24
Rank	Weighted Proposed Method Manhattan	Weighted Proposed Method Euclidean	Weighted +20% Planetary Features Method Manhattan	Weighted +20% Planetary Features Method Euclidean	Weighted -20% Planetary Features Method Manhattan	Weighted -20% Planetary Features Method Euclidean
1	Kepler-452 b	Kepler-452 b	Kepler-452 b	Kepler-1185 b	Kepler-452 b	Kepler-1253 b
2	Kepler-1638 b	Kepler-1638 b	Kepler-1185 b	Kepler-132 e	Kepler-1185 b	Kepler-1185 b
3	Kepler-51 e	Kepler-51 e	Kepler-132 e	Kepler-1253 b	Kepler-1638 b	Kepler-384 c
4	Kepler-1606 b	Kepler-1606 b	Kepler-384 c	Kepler-384 c	Kepler-1126 c	Kepler-132 e
5	Kepler-1701 b	Kepler-1701 b	Kepler-1126 c	Kepler-130 d	Kepler-132 e	Kepler-1440 b
6	Kepler-22 b	Kepler-1126 c	Kepler-1253 b	Kepler-1498 b	Kepler-384 c	Kepler-130 d
7	Kepler-1126 c	Kepler-69 c	Kepler-1965 b	Kepler-1440 b	Kepler-1253 b	Kepler-1441 b
8	Kepler-1549 b	Kepler-1185 b	Kepler-1605 b	Kepler-1620 b	Kepler-51 e	Kepler-1498 b
9	Kepler-1690 b	Kepler-442 b	Kepler-1440 b	Kepler-1441 b	Kepler-1440 b	Kepler-452 b
10	Kepler-69 c	Kepler-311 d	Kepler-1638 b	Kepler-1965 b	Kepler-1965 b	Kepler-1570 b
Top-5 overlap between weighted proposed method & unweighted ablation			1/5	0/5	2/5	0/5
Top-10 overlap between weighted proposed method & unweighted ablation			3/10	1/10	4/10	2/10

Table 4: Displays Experiment #3 Weight Sensitivity Analysis Models which compares the Weighted Proposed Method KNN Models #7 and #8 with the Weighted Sensitivity Analysis KNN Models #21, #22, #23, and #24.

Based on the results in **Table 4**, the Manhattan distance metric demonstrates noticeably higher robustness to feature-weight adjustments than the Euclidean metric. This can be seen in the Top 10 overlap produced using the Manhattan distance, which is moderately robust, sitting at 3/10 for the +20% scenario and 4/10 for the -20% scenario. The Euclidean distance, however, shows that Top 10 overlap shows minimal stability, with only 1/10 overlap for +20% weight and 2/10 overlap for -20% weight. However, both metrics show that altering planetary features significantly disrupts the baseline rankings, implying that the final recommendations are highly sensitive to how planetary mass, radius, and temperature are weighted.

Kepler-452 b remains an exceptionally robust Earth-Like candidate maintaining its ranking when planetary features are altered by $\pm 20\%$ for the Manhattan Distance. However, it is highly sensitive to the Euclidean Distance given that it drops out entirely from KNN Model #22 and drops to 9th place in KNN Model #24. Other top 5 Earth-Like candidates, Kepler-1638 b and Kepler-51 e, appear to be moderately robust appearing. Likewise, the mid-ranked Earth-Like candidates, Kepler-1126 c and Kepler-1185 b, appear to be robust.

Overall, as expected, mid-ranked candidates show a high degree of volatility for both Manhattan and Euclidean distances. For the Manhattan Distance KNN Models #21 and #23, the mid-tier candidates follow the expected sensitivity logic. While Kepler-452 b holds steady at Rank 1, Top ranked planets like Kepler-1606 b, Kepler-1701 b, are replaced by highly reactive candidates like Kepler-1185 b and Kepler-132 e, which increase to the Top 5 when planetary weights change by $\pm 20\%$ while other mid-ranked planets drop out entirely when weights shift. For the Euclidean Distance KNN Models #22 and #24, the Top 10 list undergoes a massive shift where new candidates such as Kepler-132 e and Kepler-384 c seize the Top 5 positions which were not present in the original Full KNN Models #7 and #8.

Experiment #3: Weight Sensitivity Analysis						
	Original Weights Model		Full Stellar Features +20% Weight		Full Stellar Features -20% Weight	
	Model #7	Model #8	Model #25	Model #26	Model #27	Model #28
Rank	Weighted Proposed Method Manhattan	Weighted Proposed Method Euclidean	Weighted +20% Full Stellar Features Method Manhattan	Weighted +20% Full Stellar Features Method Euclidean	Weighted -20% Full Stellar Features Method Manhattan	Weighted -20% Full Stellar Features Method Euclidean
1	Kepler-452 b	Kepler-452 b	Kepler-452 b	Kepler-1253 b	Kepler-452 b	Kepler-1185 b
2	Kepler-1638 b	Kepler-1638 b	Kepler-1185 b	Kepler-1185 b	Kepler-1185 b	Kepler-132 e
3	Kepler-51 e	Kepler-51 e	Kepler-132 e	Kepler-132 e	Kepler-132 e	Kepler-1253 b
4	Kepler-1606 b	Kepler-1606 b	Kepler-1126 c	Kepler-384 c	Kepler-1126 c	Kepler-384 c
5	Kepler-1701 b	Kepler-1701 b	Kepler-384 c	Kepler-1440 b	Kepler-384 c	Kepler-130 d
6	Kepler-22 b	Kepler-1126 c	Kepler-1638 b	Kepler-130 d	Kepler-1253 b	Kepler-1498 b
7	Kepler-1126 c	Kepler-69 c	Kepler-1253 b	Kepler-1441 b	Kepler-1638 b	Kepler-1620 b
8	Kepler-1549 b	Kepler-1185 b	Kepler-1440 b	Kepler-1498 b	Kepler-1965 b	Kepler-1440 b
9	Kepler-1690 b	Kepler-442 b	Kepler-1965 b	Kepler-1965 b	Kepler-1605 b	Kepler-442 b
10	Kepler-69 c	Kepler-311 d	Kepler-1605 b	Kepler-1505 b	Kepler-1440 b	Kepler-1965 b
Top-5 overlap between weighted proposed method & unweighted ablation			1/5	0/5	1/5	0/5
Top-10 overlap between weighted proposed method & unweighted ablation			3/10	1/10	3/10	2/10

Table 5: Displays Experiment #3 Weight Sensitivity Analysis Models which compares the Weighted Proposed Method KNN Models #7 and #8 with the Weighted Sensitivity Analysis KNN Models #25, #26, #27, and #28.

Similar to **Table 4**, the results in **Table 5** show higher robustness in results for the Manhattan Distance metric compared to the Euclidean Distance metric when all stellar feature weights are changed by $\pm 20\%$. This can be verified by the Top 5 overlap value of 1/5 and Top 10 overlap value of 3/10 for the Manhattan Distance metric while the overlap values for the Euclidean Distance metric were significantly lower with 0 Top 5 overlap and 1-2 Top 10 overlap.

Additionally, the Kepler-452 b once again remains highly robust for the Manhattan Distance KNN Models #25 and #27 but is no longer present in the Top 10 list for the Euclidean Distance KNN Models #26 and #28 due to its high sensitivity. Top 5 ranked planet Kepler-1638 b and mid-ranked planet Kepler-1126 c remain moderately robust for Manhattan Distance KNN Models #25 and #27 where both planets increase in ranking after weight change compared to the ranking from the original Full KNN Models #7 and #8. Mid-ranked planet Kepler-1185 b appears to be moderately robust as its ranking increases after weight shifts while the mid-ranked planet Kepler-442 b appears to be weakly robust showing consistent ranking when stellar feature weights decrease by 20% but dropping out of list when stellar feature weights increase by 20%. All other top ranked and mid-ranked planets from the Original Weights Model remained highly volatile for both distance metrics.

Experiment #3: Weight Sensitivity Analysis						
	Original Weights Model		Partial Stellar Features +20% Weight		Partial Stellar Features -20% Weight	
	Model #7	Model #8	Model #29	Model #30	Model #31	Model #32
Rank	Weighted Proposed Method Manhattan	Weighted Proposed Method Euclidean	Weighted +20% Partial Stellar Features Method Manhattan	Weighted +20% Partial Stellar Features Method Euclidean	Weighted -20% Partial Stellar Features Method Manhattan	Weighted -20% Partial Stellar Features Method Euclidean
1	Kepler-452 b	Kepler-452 b	Kepler-452 b	Kepler-1253 b	Kepler-452 b	Kepler-1185 b
2	Kepler-1638 b	Kepler-1638 b	Kepler-1185 b	Kepler-1185 b	Kepler-1185 b	Kepler-132 e
3	Kepler-51 e	Kepler-51 e	Kepler-132 e	Kepler-132 e	Kepler-132 e	Kepler-1253 b
4	Kepler-1606 b	Kepler-1606 b	Kepler-1126 c	Kepler-384 c	Kepler-1126 c	Kepler-384 c
5	Kepler-1701 b	Kepler-1701 b	Kepler-384 c	Kepler-1440 b	Kepler-384 c	Kepler-130 d
6	Kepler-22 b	Kepler-1126 c	Kepler-1638 b	Kepler-130 d	Kepler-1253 b	Kepler-1498 b
7	Kepler-1126 c	Kepler-69 c	Kepler-1253 b	Kepler-1441 b	Kepler-1638 b	Kepler-1440 b
8	Kepler-1549 b	Kepler-1185 b	Kepler-1440 b	Kepler-1498 b	Kepler-1965 b	Kepler-1965 b
9	Kepler-1690 b	Kepler-442 b	Kepler-1965 b	Kepler-1620 b	Kepler-1605 b	Kepler-1441 b
10	Kepler-69 c	Kepler-311 d	Kepler-1605 b	Kepler-1965 b	Kepler-1440 b	Kepler-1620 b
Top-5 overlap between weighted proposed method & unweighted ablation			1/5	0/5	1/5	0/5
Top-10 overlap between weighted proposed method & unweighted ablation			3/10	1/10	3/10	1/10

Table 6: Displays Experiment #3 Weight Sensitivity Analysis Models which compares the Weighted Proposed Method KNN Models #7 and #8 with the Weighted Sensitivity Analysis KNN Models #29, #30, #31, and #32.

Experiment #3: Weight Sensitivity Analysis						
	Original Weights Model		Spectral Type Features +20% Weight		Spectral Type Features -20% Weight	
	Model #7	Model #8	Model #33	Model #34	Model #35	Model #36
Rank	Weighted Proposed Method Manhattan	Weighted Proposed Method Euclidean	Weighted +20% Spectral Type Features Method Manhattan	Weighted +20% Spectral Type Features Method Euclidean	Weighted -20% Spectral Type Features Method Manhattan	Weighted -20% Spectral Type Features Method Euclidean
1	Kepler-452 b	Kepler-452 b	Kepler-452 b	Kepler-1185 b	Kepler-452 b	Kepler-1185 b
2	Kepler-1638 b	Kepler-1638 b	Kepler-1185 b	Kepler-1253 b	Kepler-1185 b	Kepler-1253 b
3	Kepler-51 e	Kepler-51 e	Kepler-132 e	Kepler-132 e	Kepler-132 e	Kepler-132 e
4	Kepler-1606 b	Kepler-1606 b	Kepler-1126 c	Kepler-384 c	Kepler-1126 c	Kepler-384 c
5	Kepler-1701 b	Kepler-1701 b	Kepler-384 c	Kepler-130 d	Kepler-384 c	Kepler-130 d
6	Kepler-22 b	Kepler-1126 c	Kepler-1638 b	Kepler-1440 b	Kepler-1638 b	Kepler-1440 b
7	Kepler-1126 c	Kepler-69 c	Kepler-1253 b	Kepler-1498 b	Kepler-1253 b	Kepler-1498 b
8	Kepler-1549 b	Kepler-1185 b	Kepler-1965 b	Kepler-1441 b	Kepler-1965 b	Kepler-1620 b
9	Kepler-1690 b	Kepler-442 b	Kepler-1440 b	Kepler-1965 b	Kepler-1440 b	Kepler-1441 b
10	Kepler-69 c	Kepler-311 d	Kepler-1605 b	Kepler-1505 b	Kepler-1605 b	Kepler-1965 b
Top-5 overlap between weighted proposed method & unweighted ablation			1/5	0/5	1/5	0/5
Top-10 overlap between weighted proposed method & unweighted ablation			3/10	1/10	3/10	1/10

Table 7: Displays Experiment #3 Weight Sensitivity Analysis Models which compares the Weighted Proposed Method KNN Models #7 and #8 with the Weighted Sensitivity Analysis KNN Models #33, #34, #35, and #36.

The results in **Table 6** and **Table 7** both show that Manhattan Distance KNN models #33 and #35 display a higher robustness to weight changes compared to the Euclidean Distance KNN models #34 and #36. Kepler-452 b remained highly stable while Kepler-1638 b and Kepler-1126 c remained moderately robust for the Manhattan Distance KNN models #33 and #35. Kepler-1185 b remained moderately robust for the Euclidean Distance KNN models #34 and #36. All other top-ranked exoplanets and mid-ranked exoplanets from the Original Weights Model were unstable in their rankings when part of the stellar property weights or only the spectral type feature weights were changed by $\pm 20\%$.

Experiment #3: Weight Sensitivity Analysis						
	Original Weights Model		Orbital and HZ Features +20% Weight		Orbital and HZ Features -20% Weight	
	Model #7	Model #8	Model #37	Model #38	Model #39	Model #40
Rank	Weighted Proposed Method Manhattan	Weighted Proposed Method Euclidean	Weighted +20% Orbital and Habitable Zone (HZ) Features Method Manhattan	Weighted +20% Orbital and Habitable Zone (HZ) Features Method Euclidean	Weighted -20% Orbital and Habitable Zone (HZ) Features Method Manhattan	Weighted -20% Orbital and Habitable Zone (HZ) Features Method Euclidean
1	Kepler-452 b	Kepler-452 b	Kepler-452 b	Kepler-1185 b	Kepler-452 b	Kepler-1185 b
2	Kepler-1638 b	Kepler-1638 b	Kepler-1185 b	Kepler-1253 b	Kepler-1185 b	Kepler-1253 b
3	Kepler-51 e	Kepler-51 e	Kepler-1638 b	Kepler-132 e	Kepler-132 e	Kepler-132 e
4	Kepler-1606 b	Kepler-1606 b	Kepler-1126 c	Kepler-384 c	Kepler-384 c	Kepler-384 c
5	Kepler-1701 b	Kepler-1701 b	Kepler-132 e	Kepler-130 d	Kepler-1126 c	Kepler-130 d
6	Kepler-22 b	Kepler-1126 c	Kepler-384 c	Kepler-1440 b	Kepler-1253 b	Kepler-1440 b
7	Kepler-1126 c	Kepler-69 c	Kepler-1253 b	Kepler-1498 b	Kepler-1965 b	Kepler-1498 b
8	Kepler-1549 b	Kepler-1185 b	Kepler-51 e	Kepler-1441 b	Kepler-1440 b	Kepler-1441 b
9	Kepler-1690 b	Kepler-442 b	Kepler-1965 b	Kepler-452 b	Kepler-1605 b	Kepler-1965 b
10	Kepler-69 c	Kepler-311 d	Kepler-1440 b	Kepler-1620 b	Kepler-1638 b	Kepler-1620 b
Top-5 overlap between weighted proposed method & unweighted ablation			2/5	0/5	1/5	0/5
Top-10 overlap between weighted proposed method & unweighted ablation			4/10	2/10	3/10	1/10

Table 8: Displays Experiment #3 Weight Sensitivity Analysis Models which compares the Weighted Proposed Method KNN Models #7 and #8 with the Weighted Sensitivity Analysis KNN Models #37, #38, #39, and #40.

When altering the orbital and habitable zone feature weights by $\pm 20\%$, the results from **Table 8** mirror the results obtained from **Table 4** which show that Kepler-452 b remains stable in its ranking for the Manhattan Distance KNN models #37 and #39 while its ranking dropped to 9th place for the Euclidean Distance KNN Model #38. Therefore, Kepler-452 b is robust for the Manhattan Distance metric but not for the Euclidean Distance metric. Kepler-1638 b and Kepler-1126 c remained moderately stable while Kepler-51 e was weakly stable for the Manhattan Distance metrics. Kepler-1185 b remained moderately stable for the Euclidean Distance metrics. However, all other top-ranked planets and min-ranked planets for both distance metrics were unstable when compared to the Original Weights Model Top 10 planet list.

Overall, the results from Experiment #4 show that when weights for key features change, Kepler-452 b is highly robust but only when using the Manhattan distance metric but fails to maintain its ranking when using the Euclidean distance metric due to its higher feature similarity sensitivity compared to the Manhattan distance metric. Given that Kepler-452 b demonstrated the highest-ranking robustness across weight perturbation models, this suggests that its Earth-like similarity is intrinsically multi-dimensional rather than dependent on

specific feature-weight prioritization choices. Kepler-1638 b, Kepler-1126 c, Kepler-1185 b, and occasionally Kepler-51 e and Kepler-442 b appeared across the top planet rankings produced from the various weight scenarios. However, their rankings shifted compared to the models from the original proposed weight settings. Therefore, making them moderately robust. All other top-ranked and mid-ranked planets showed major shifts showing overall weak robustness. Therefore, feature-weight choices meaningfully influence candidate prioritization.

Experiment #4: Imputation Sensitivity Analysis

The purpose of this experiment is to determine whether the top 10 Earth-Like planet rankings depend too heavily on imputed values. Since several important exoplanet features had missing values before imputation, we compared the current final imputed dataset with a more conservative version of the data. The Full Imputed Dataset Model is represented by the Weighted Proposed Method Manhattan KNN Model #7 and Weighted Proposed Method Euclidean KNN Model #8.

The goal is not necessarily to get identical rankings but rather to show how much the rankings depend on imputation assumptions. If the strongest candidates remain stable, the top-ranked results are said to be reasonably robust. If the lower-ranked candidates change, borderline candidates should be interpreted more cautiously since public exoplanet catalogs contain incomplete measurements.

The Full Imputed Dataset Model was compared to the Reduced Feature Dataset Model which is represented by the Weighted Reduced Features Method Manhattan KNN Model #41 and Weighted Reduced Features Method Euclidean KNN Model #42. The Reduced Feature Model removed highly imputed features which originally had at least 30% missing data and the models were rerun with the remaining features. These features used the same proposed method weight setting from KNN Models #7 and #8.

The key features removed were spectral type and planet mass which has missing values of 68.57% and 47.62% respectively. After these features were removed, the remaining key features included planet radius, planet equilibrium temperature, orbital period, orbital semi major axis, number of stars in the system, stellar surface gravity, stellar radius, stellar mass, and stellar effective temperature. Missing data present for these features were imputed using relevant astronomical formulas similar to the Full Imputed Dataset Model where planet name corresponds to the observation.

However, since planet mass was initially removed due to its large percentage of missing data, any remaining missing data for planet radius could not be imputed for which relies on the presence of planet mass.

Therefore, 24.91% of missing data was present for planet radius. Additionally, stellar radius and stellar mass missing values cannot be imputed since no current astronomical formulas are available to estimate to impute for them. Hence, 9.69% missing stellar radius and 0.41% missing stellar mass data. As a result, these rows with missing data were removed which resulted in a total of 4,525 exoplanets. Lastly, while the planetary orbital velocity and stellar luminosity features could be created, the habitable-zone flag feature could not be due to the removal of the spectral type column which is used to determine the habitable zone range. Therefore, only 12 key features remained rather than the original 15 key features from the Full Imputed Dataset Model.

Another version of the Reduced Feature Dataset Model called the Reduced Feature Dataset (No Planet Radius) Model, was created with planet radius being removed to determine. This is represented by the Weighted Reduced Features (No Planet Radius) Method Manhattan KNN Model #43 and Weighted Reduced Features (No Planet Radius) Method Euclidean KNN Model #44. So, any remaining rows with missing data corresponding to only missing stellar radius and mass data were removed. Therefore, with less rows containing missing data, this resulted in 5,437 with 11 key features remaining.

Additionally, the Full Imputed Dataset Model was compared to the Complete Case Dataset Model which filtered for rows in the dataset corresponding to exoplanets that originally had complete values for the key ranking features before imputation. Therefore, any exoplanet which consisted of at least one key feature with missing data was removed. This resulted in 758 exoplanets with all 15 key features remaining for this model. The Complete Case Dataset Model represented by the Weighted Complete Case Dataset Model Method Manhattan KNN Model #45 and Weighted Complete Case Dataset Model Method Euclidean KNN Model #46. These features in the Complete Case Dataset Model used the same proposed method weight setting from KNN Models #7 and #8.

Experiment #4: Imputation Sensitivity Analysis								
	Full Imputed Dataset Model		Reduced Feature Dataset Model		Reduced Feature Dataset (No Planet Radius) Model		Complete Case Dataset Model	
	Model #7	Model #8	Model #41	Model #42	Model #43	Model #44	Model #45	Model #46
Rank	Weighted Proposed Method Manhattan	Weighted Proposed Method Euclidean	Weighted Reduced Features Method Manhattan	Weighted Reduced Features Method Euclidean	Weighted Reduced Features (No Planet Radius) Method Manhattan	Weighted Reduced Features (No Planet Radius) Method Euclidean	Weighted Complete Case Dataset Model Method Manhattan	Weighted Complete Case Dataset Model Method Euclidean
1	Kepler-452 b	Kepler-452 b	Kepler-452 b	Kepler-452 b	Kepler-139 f	Kepler-1634 b	Kepler-22 b	Kepler-22 b
2	Kepler-1638 b	Kepler-1638 b	Kepler-62 f	Kepler-62 f	Kepler-452 b	HD 100777 b	HD 224018 d	HD 224018 d
3	Kepler-51 e	Kepler-51 e	Kepler-22 b	Kepler-1126 c	HD 28185 b	Kepler-139 f	K2-3 d	K2-3 d
4	Kepler-1606 b	Kepler-1606 b	Kepler-1638 b	Kepler-1638 b	HD 100777 b	HD 108874 b	HD 136352 d	Kepler-138 d
5	Kepler-1701 b	Kepler-1701 b	Kepler-69 c	Kepler-69 c	HD 142415 b	Kepler-452 b	TRAPPIST-1 d	K2-3 c
6	Kepler-22 b	Kepler-1126 c	Kepler-1632 b	Kepler-441 b	HD 108874 b	HD 92788 b	Kepler-138 d	K2-136 d
7	Kepler-1126 c	Kepler-69 c	Kepler-1126 c	Kepler-1544 b	Kepler-1634 b	HD 43197 b	Gliese 12 b	TRAPPIST-1 d
8	Kepler-1549 b	Kepler-1185 b	Kepler-311 d	Kepler-311 d	HD 92788 b	HD 28185 b	LP 791-18 d	Gliese 12 b
9	Kepler-1690 b	Kepler-442 b	Kepler-289 e	Kepler-22 b	HD 43197 b	HD 142415 b	K2-138 g	TOI-2095 b
10	Kepler-69 c	Kepler-311 d	Kepler-1544 b	Kepler-1632 b	Kepler-289 e	Kepler-289 e	K2-3 c	Kepler-138 c
Top-5 overlap between weighted proposed method & unweighted ablation			2/5	2/5	1/5	1/5	1/5	0/5
Top-10 overlap between weighted proposed method & unweighted ablation			5/10	5/10	1/10	1/10	1/10	0/10

Table 9: Displays Experiment #3 Weight Sensitivity Analysis Models which compares the Weighted Proposed Method KNN Models #7 and #8 with the Weighted Sensitivity Analysis KNN Models #41, #42, #43, and #44.

The results in **Table 9**, comparing the Full Imputed Dataset Model with the Reduced Feature Dataset Model reveal that Kepler-452 b remains highly robust for both the Manhattan and Euclidean distance metrics remaining as the top Earth-Like candidate when high imputed features are removed. The Top 5 ranked Earth-Like candidate, Kepler-1638 b, also remains robust for both Manhattan and Euclidean distance metrics though its ranking dropped from 2nd place to 4th place for both metrics. Mid-ranked Earth-Like candidates, Kepler-22 b, and Kepler-1126 c, appeared somewhat robust for the Reduced Feature Dataset Model. Kepler-22 b's ranking increased for both distance metrics while Kepler-1126 c remained in the same ranking for the Manhattan distance metric and increased for the Euclidean distance metric. Other mid-ranked candidates, Kepler-69 c and Kepler-311 d also appeared to be somewhat robust as its rankings increased for the Manhattan distance metric and Euclidean distance metric respectively. All other top and mid ranked candidates were not present for both distance metrics.

Comparing the Full Imputed Dataset Model with the Reduced Feature (No Planet Radius) Dataset Model reveals that Kepler-452 b continues to be reasonably robust as it remains in the top 5 list though its

ranking drops slightly for both distance metrics. No other top and mid ranked candidates were present for both distance metrics which reveals the unique composition of Kepler-452 b's similarity to Earth.

The greatest changes were visible when comparing the Full Imputed Dataset Model with the Complete Case Dataset Model which showed that only Kepler-22 b remained in the top 10 list for the Manhattan distance metric only, but no other top 10 candidates were present for the Manhattan distance. The Euclidean distance metric displayed the biggest change as none of the top 10 planets from the Full Imputed Dataset Model were present for the Euclidean distance. These results can be expected since a majority of the top 10 Earth-Like candidates produced from the Full Imputed Dataset Model contained at least one missing property which had to be imputed.

Overall, the results from Experiment #4 show that the highest-ranked Earth-Like candidates produced from the Full Imputed Dataset Model demonstrate varying levels of robustness when tested against datasets with reduced dependence on imputed values. Kepler-452 b remained the most consistently robust candidate across nearly all dataset variations and for both the Manhattan and Euclidean distance metrics, indicating that its Earth-like similarity is not heavily dependent on imputation assumptions or any single highly imputed feature. Kepler-1638 b, Kepler-22 b, Kepler-1126 c, Kepler-69 c, and Kepler-311 d also displayed moderate robustness since they continued to appear within several reduced-feature ranking models, although their rankings shifted compared to the Full Imputed Dataset Model. However, when highly imputed features such as planet mass, spectral type, and eventually planet radius were removed, the majority of the remaining top-ranked candidates changed substantially, particularly for the Euclidean distance metric. The Complete Case Dataset Model produced the greatest ranking instability, especially under the Euclidean distance metric where none of the original top 10 candidates remained present. These findings suggest that while a small subset of planets, particularly Kepler-452 b, possess stable multi-feature Earth-like characteristics, many other candidate rankings remain sensitive to incomplete measurements and imputation assumptions present in current public exoplanet catalogs.

Experiment #5: Literature/Catalog Comparison

The purpose of this experiment is to compare our top-ranked Earth-Like exoplanets produced from both the Manhattan and Euclidean Distance KNN Models #1 and #2 with known exoplanet catalogs and literature discussions of potentially habitable or Earth-like exoplanets. We recorded whether a particular exoplanet from our top-ranked list has been previously discussed as Earth-like, potentially habitable, located in or near the habitable zone, or is notable for any habitability-related reason. The goal is to determine if our model results are reasonable and literature-consistent candidates, rather than purely random outcomes. The catalogs used for this comparison include the NASA Exoplanet Archive [16], NASA Exoplanet Catalog [33], Habitable Worlds Catalog by the Planetary Habitability Laboratory [29], The Extrasolar Planets Encyclopaedia [45], and Open Exoplanet Catalogue [46]. The various literature used for this comparison can be referenced from [32], [34]-[44], and [46]-[71]. The results of this experiment can be referenced in **Table 10**.

Experiment #5: Literature/Catalog Comparison				
Planet	Rank in Our Manhattan Distance KNN Model	NASA/Literature Relevance	Habitability-Related Notes	Limitation or Caution
Kepler-452 b	1	Previously discussed as Earth-like candidate according to NASA Exoplanet Archive, NASA Exoplanet Catalog, and other literature. [16][32][33][34][37][55] Currently only exoplanet coined as "Earth-cousin" [33] Orbital period of 384.843 days. Orbital distance 1.046 AU. Planet equilibrium temperature of 265 K. Stellar effective temperature of 5757 K. Stellar radius of 1.11Rs. Stellar mass of 1.037 Ms. Super-Earth Like in radius at 1.63Rp. [16][29][32][33][34][36][41][45][46][47][60][62][63] Orbits a host star similar to the Sun in terms of age and metallicity. [16][38][46] Super-Earth Like in mass at ~3.29Mp.[33]. Super-Earth Like in mass at ~5Mp.[32]. Super-Earth Like in mass at ~6Mp.[35][46] Has an Earth Similarity Index (ESI) of 0.83 [29][39][40]	Orbits and is in or near the Habitable Zone (HZ) of a Main Sequence Sun-Like Star (GV). [16][29][32][33][34][35][39][40][45][46][55][63][68] Likely has always been in the habitable zone and should remain there for another ~3 Gyr. [32][34] Within the Photosynthetic Habitable Zone (PHZ) which is the distance from a star where both liquid water and oxygenic photosynthesis can occur. [38] Ranked in 19th place for potentially habitable exoplanets by the Habitable Worlds Catalog (HWC) sorted by the Earth Similarity Index (ESI). [29] Ranked in 25th place for habitability potential by Biological Timescale Habitability Score (BTHS) [47] Has a Planetary Habitability Index (PHI) score of 0.89 and a Statistical-likelihood Exo-Planetary Habitability Index (SEPHI) of 0.81. [68] Referenced as a potentially water rich rocky habitable planet. [33][41]	Experiences 10% more flux (radiation) from its star than Earth receives from the Sun. [32][39] Planet mass of ~5Mp is approximated by NASA (not measured) The likelihood that this planet has a rocky composition lies between 49% and 62%. [32][34] Uncertainty of planet composition. Likely rocky planet but one source claims it could be gaseous or a Waterworld. [35] Kepler-452 b could experience a runaway greenhouse effect. [33][34][35]

Kepler-1638 b	2	Orbital period of 259.3 days. Orbital distance of 0.745 AU. Planet equilibrium temperature of 328 K. Stellar effective temperature of 5737 K. Stellar radius of 1.3Rs. Stellar mass of 0.91 Ms. Super-Earth Liker radius of 1.87Rp [16][29][33][39][40][41][42][45][46][56][60][63] Super-Earth Like mass of ~4.16Mp. [33][39][40][46] Orbits a host star similar to the Sun in terms of mass, temperature, age, and metallicity. [16][39][40][45][46] There could be an atmosphere based on the presence of carbon dioxide [39][40] Has an Earth Similarity Index (ESI) of 0.76 [29][39][40]	Orbits a Main Sequence Type G star (GV) within or near its habitable zone. [16][33][39][40][45][46][29][63] Referenced as a potentially water rich rocky habitable planet. [33][41][42] Within the Photosynthetic Habitable Zone (PHZ) which is the distance from a star where both liquid water and oxygenic photosynthesis can occur. [38] Ranked in 30th place for potentially habitable exoplanets by the Habitable Worlds Catalog (HWC) sorted by the Earth Similarity Index (ESI). [29]	Super-Earth Like planet mass of ~4.16Mp is approximated by NASA (not measured). Short orbital period not in range of 300-400 days. Uncertainty of planet composition due to missing mass data. Likely rocky planet but could be gaseous.
Kepler-51 e	3	Orbital period of 264 days. Super-Earth Like in mass at 4.6-5.4Mp. [16][33][43][45] Stellar effective temperature of 5674 K. Stellar radius of 0.87Rs. Stellar mass of 0.96Ms. [16][33][43][44][45] Neptune Like in radius at ~2.18Rp. [33]	Orbits a Main Sequence Type G star (GV) just in or near its habitable zone. [16][33][45]	Classified as a Neptune-Like planet. [33] Low planet density. Known as a "super puff" planet. [43] Short orbital period not in range of 300-400 days. Orbital distance and planet equilibrium temperature are unknown (were imputed properties) Super-Earth Like radius of ~2.18Rp approximated by NASA (not measured)
Kepler-1606 b	4	Orbital period of 196.4 days. Orbital distance of 0.642 AU. Neptune-Like radius of 2.07 Rp. Planet equilibrium temperature 288 K. Stellar effective temperature 5461 K. Stellar radius of 0.83Rs. Stellar mass of 0.915Ms. [16][29][33][37][45][46][47][48][56][63] Super-Earth Like mass of ~4.49Mp. [33][29][46] Has an Earth Similarity Index (ESI) of 0.70 [29][48]	Orbits a Main Sequence Type G star (GV) within or near its habitable zone. [16][33][37][45][46][29][48][63] Referenced as a potentially rocky planet [33] Ranked in 11th place for habitability potential by Biological Timescale Habitability Score (BTHS) [47] Ranked in 43rd place for potentially habitable exoplanets by the Habitable Worlds Catalog (HWC) sorted by the Earth Similarity Index (ESI). [29]	Super-Earth Like planet mass of ~4.49Mp is approximated by NASA (not measured). Short orbital period not in range of 300-400 days. Uncertainty of planet composition due to missing mass data and Neptune-Like size. Thought likely Super-Earth Like planet.

Kepler-1701 b	5	Orbital period of 169 days. Orbital distance of 0.56 AU. Neptune-Like radius of 2.2Rp. Planet equilibrium temperature of 276 K. Stellar effective temperature of 5250 K. Stellar radius of 0.794Rs. Stellar mass of 0.867Ms. [16][29][33][37][45][46][47][48][63] Super-Earth Like mass of ~5.57Mp [33] Has an Earth Similarity Index (ESI) of 0.71 [29][48]	Orbits a Main Sequence Type K star (KV) within or near its habitable zone. [16][33][37][29][45][46][48][63] Ranked in 20th place for habitability potential by Biological Timescale Habitability Score (BTHS) [47] Ranked in 41st place for potentially habitable exoplanets by the Habitable Worlds Catalog (HWC) sorted by the Earth Similarity Index (ESI). [29]	Classified as a Neptune-Like planet. [33] Does not orbit a Type GV host star. Super-Earth Like planet mass of ~5.57Mp is approximated by NASA (not measured). Short orbital period not in range of 300-400 days. Uncertainty of planet composition due to missing atmosphere data. [72]
Kepler-22 b	6	Orbital period of 289 days. Orbital distance of 0.83 AU. Neptune-Like radius of 2.1-2.4Rp. Planet equilibrium temperature of 264 K. Stellar effective temperature of 5589 K. Stellar radius of 0.88Rs. Stellar mass of 0.95Ms. [16][29][33][37][45][46][49][50][51][52][53][54][55][62][63] Neptune-Like planet mass of 9.1Mp. [16][29][33][37][45][46][50][51][52][53][54][55] Neptune-Like planet mass of 36Mp, 82Mp, 124Mp [49] Has an Earth Similarity Index (ESI) of 0.72 [29]	Orbits a Main Sequence Type G star (GV) within or near its habitable zone. [16][33][37][29][45][46][49][50][51][52][53][54][55][63][68] Ranked in 37th place for potentially habitable exoplanets by the Habitable Worlds Catalog (HWC) sorted by the Earth Similarity Index (ESI). [29] Has a Planetary Habitability Index (PHI) score of 0.66 and an Statistical-likelihood Exo-Planetary Habitability Index (SEPHI) of 0.83. [68] Referenced as a potentially water rich rocky habitable planet. [33][51][53][54]	Neptune-Like planet mass of 9.1Mp is currently used as default value by NASA but its planet mass has a range of different values: 36Mp, 82Mp, and 124Mp [49] Short orbital period not in range of 300-400 days. Uncertainty of planet composition. Could be rocky, gaseous or liquid composition. [73]
Kepler-1126 c	7	Orbital period of 199.7 days. Orbital distance of 0.62 AU. Earth-Like radius of 1.45-1.54Rp. Planet equilibrium temperature of 305 K. Stellar effective temperature of 5678 K. Stellar radius of 0.92Rs. Stellar mass of 0.79Ms. [16][29][33][37][46][47][57][60] Super-Earth Like planet mass of ~2.7Mp. [33] Has an Earth Similarity Index (ESI) of 0.72 [29]	Orbits a Main Sequence Type G star (GV) within or near its habitable zone. [16][29][33][37][46][57] Ranked in 15th place for habitability potential by Biological Timescale Habitability Score (BTHS) [47] Ranked in 49th place for potentially habitable exoplanets by the Habitable Worlds Catalog (HWC) sorted by the Earth Similarity Index (ESI). [29] Referenced as a moist greenhouse planet candidate. [57] Referenced as a potentially rocky planet [33]	Super-Earth Like planet mass of ~2.7Mp is approximate by NASA (not measured). Limitations in determining the composition of potentially habitability of planets. [57] Short orbital period not in range of 300-400 days.

Kepler-1549 b	8	Orbital period of 214 days. Orbital distance of 0.67 AU. Neptune-Like radius of 2.61 Rp. Planet equilibrium temperature of 271 K. Stellar effective temperature of 5282 K. Stellar radius of 0.885 Rs. Stellar mass of 0.87Ms. [16][29][33][37][45][46][62][63] Super-Earth Like mass of ~7.1Mp. [33] Has an Earth Similarity Index (ESI) of 0.68 [29]	Orbits a Main Sequence Type G star (GV) within or near its habitable zone. [16][29][33][37][45][46][63] Ranked in 70th place for potentially habitable exoplanets by the Habitable Worlds Catalog (HWC) sorted by the Earth Similarity Index (ESI). [29]	Classified as a Neptune-Like planet [33] Super-Earth Like planet mass of ~7.1Mp is approximated by NASA (not measured). Short orbital period not in range of 300-400 days.
Kepler-1690 b	9	Orbital period of 234..6 days. Orbital distance of 0.7097 AU. Neptune-Like radius of 2.81Rp. Planet equilibrium temperature of 276 K. Stellar effective temperature of 5589.5 K. Stellar radius of 0.851Rs. Stellar mass of 0.89Ms. [16][29][33][45][46][62][63] Super-Earth Like mass of ~8.3Mp. [33] Has an Earth Similarity Index (ESI) of 0.66 [29]	Orbits a Main Sequence Type G star (GV) within or near its habitable zone. [16][29][33][45][46][62][63] Ranked in 84th place for potentially habitable exoplanets by the Habitable Worlds Catalog (HWC) sorted by the Earth Similarity Index (ESI). [29]	Classified as a Neptune-Like planet [33] Super-Earth Like planet mass of ~8.3Mp is approximated by NASA (not measured). There is speculation about its potential habitability. Short orbital period not in range of 300-400 days.
Kepler-69 c	10	Orbital period of 242.5 days. Orbital distance of 0.64-0.71 AU. Super-Earth-Like radius of 1.71-1.84Rp. Planet equilibrium temperature of 292 K. Stellar effective temperature of 5696 K. Stellar radius of 0.95Rs. Stellar mass of 0.88Ms. [16][29][33][45][46][52][56][59][63][64][65][66] Super-Earth Like mass of ~3.57Mp. [33] Has an Earth Similarity Index (ESI) of 0.70 [29]	Orbits a Main Sequence Type G star (GV) within or near its habitable zone. [16][29][33][45][46][52][56][59][63][64][65][66][67][68] Has a Planetary Habitability Index (PHI) score of 0.79 and an Statistical-likelihood Exo-Planetary Habitability Index (SEPHI) of 0.87. [68] Ranked in 58th place for potentially habitable exoplanets by the Habitable Worlds Catalog (HWC) sorted by the Earth Similarity Index (ESI). [29] Referenced as a potentially rocky planet [33]	May not be a planet of interest in future missions [59] Super-Earth Like planet mass of ~3.57Mp is approximated by NASA (not measured). Could be classified as a Super-Venus [66]. While other sources classify it as a Super-Earth planet [33] There is speculation about its potential habitability. Short orbital period not in range of 300-400 days.
Planet	Rank in Our Euclidean Distance KNN Model	NASA/Literature Relevance	Habitability-Related Notes	Limitation or Caution
Kepler-1185 b	8	Orbital period of 104.4 days. Orbital distance of 0.43 AU. Earth-Like/Super-Earth-Like radius of 1.33-1.56Rp. Planet equilibrium temperature of 365.7 K. Stellar effective temperature of 5738 K. Stellar radius of 0.92Rs. Stellar mass of 1.04Ms [16][33][45][46]	Orbits a Main Sequence Type G star (GV) [16][33][45][46] Could possibly within or near its star's habitable zone. Referenced as a potentially rocky planet [33]	Super-Earth Like planet mass of ~2.33Mp is approximated by NASA (not measured). Habitability potential is uncertain.

		Super-Earth Like mass of ~2.33Mp. [33] Has an Earth Similarity Index (ESI) of 0.41 [29]	Ranked in 443rd place for potentially habitable exoplanets by the Habitable Worlds Catalog (HWC) sorted by the Earth Similarity Index (ESI). [29]	Short orbital period not in range of 300-400 days.
Kepler-442 b	9	Previously discussed as Earth-like candidate according to NASA Exoplanet Archive, NASA Exoplanet Catalog, and other literature.[16][33] Orbital period of 112.3 days. Orbital distance of 0.4 AU. Earth-Like radius of 1.34-1.47Rp. Planet equilibrium temperature of 255 K. Stellar effective temperature of 4510 K. Stellar radius of 0.63Rs. Stellar mass of 0.65Ms [29][37][39][40][45][46][47][52][54][57][61][63][69][70][71] Super-Earth Like mass of ~2.36Mp. [33] Has an Earth Similarity Index (ESI) of 0.84 [29][39][40]	Orbits a Main Sequence Type K star (KV) within or near its habitable zone.[29][37][39][40][45][46][52][54][57][63][68][70][71] Ranked in 5th place for habitability potential by Biological Timescale Habitability Score (BTHS) [47] Ranked in 17th place for potentially habitable exoplanets by the Habitable Worlds Catalog (HWC) sorted by the Earth Similarity Index (ESI).[29] Has a Planetary Habitability Index (PHI) score of 0.90 and an Statistical-likelihood Exo-Planetary Habitability Index (SEPHI) of 0.94. [68] Referenced as a moist greenhouse planet candidate. [57] Referenced as a potentially water rich rocky habitable planet. [33][70] Referenced to have a habitability index even higher than that of Earth. and has a Habitable Index for Transiting Planets (HITE) score higher than Earth at 0.829. [39][40]	Does not orbit a Type GV host star. Super-Earth Like planet mass of ~2.36Mp is approximated by NASA (not measured). Short orbital period not in range of 300-400 days. Planet atmosphere and surface composition are unknown.
Kepler-311 d	10	Orbital period of 232 days. Orbital distance of 0.73 AU. Super-Earth-Like/sub-Neptune-Like radius of 2.05Rp. Planet equilibrium temperature of 319.7 K. Stellar effective temperature of 5889.7 K. Stellar radius of 1.15Rs. Stellar mass of 0.98Ms. [16][33][46][58][60] Super-Earth Like mass of ~4.86Mp. [33] Has an Earth Similarity Index (ESI) of 0.58 [29]	Orbits a Main Sequence Type G star (GV) [16][33][46] Could possibly within or near its star's habitable zone. Referenced as a potentially rocky planet [33] Ranked in 159th place for potentially habitable exoplanets by the Habitable Worlds Catalog (HWC) sorted by the Earth Similarity Index (ESI). [29]	Super-Earth Like planet mass of ~4.86Mp is approximated by NASA (not measured). Habitability potential is uncertain. Short orbital period not in range of 300-400 days.

Table 10: Displays Experiment #5: Literature/Catalog Comparison which compares the top 10 Earth-Like exoplanets from the Weighted Proposed Method KNN Models #1 and #2 with other catalogs and literature (Note: the top planets ranked 1-7 were not displayed for the Euclidean KNN Model #2 to avoid redundancy with the Manhattan KNN Model #1).

The results from Experiment #5 demonstrate that both KNN models produce exoplanet rankings that are broadly consistent with established exoplanet habitability literature and catalogs. A majority of the top 10 Earth-Like candidates found from our models are independently flagged as “potentially habitable” or “Earth-like candidates” falling in or near the habitable zones of their host stars. These planets tended to orbit a Main Sequence Type-G star with only two planets orbiting Main Sequence Type-K stars. These exoplanets were Super-Earth Like or sub-Neptune-Like in mass, size, or a combination of both. A majority of the exoplanets tend to have warmer equilibrium temperatures compared to Earth with significantly shorter orbital periods. Additionally, a sizable portion of these exoplanets had orbital distances between 0.6-0.8 AU. This shows that our models are not random artifacts and can display real astrophysical habitability patterns.

Out of all the exoplanets in our top-ranked list, Kepler-452 b, Kepler-1638 b, Kepler-1606 b, Kepler-22 b, Kepler-1126 c, and Kepler-442 b appear to have the highest likelihood of being potentially habitable rocky planets based on agreement with established literature and catalog listings. Out of which, Kepler-452 b and Kepler-442 b appear to have the greatest probability of being potentially habitable Earth-Like candidates as their rankings align with independently developed habitability indices (ESI, PHI, HWC rankings). While Kepler-1126 c and Kepler-22 b are classified as Neptune-Like in mass, radius, and potentially surface composition, they exhibit notable Earth-Like similarities in terms of combined similarity with planetary equilibrium temperature, orbital period, and orbital distance with Sun-Like stellar properties.

Kepler-452 b remains to have the highest likelihood of being an Earth-Like candidate based on its planetary feature similarity with Earth and its host star’s stellar similarity to the Sun. This is particularly evident in its combined similarity in orbital period, orbital distance, and equilibrium temperature to that of Earth’s which explains its unique composition in relation to Earth compared to other planets. This is consistent with prior catalog and literature discussions.

However, the results from this experience also highlight significant astrophysical uncertainties, particularly in planetary composition and mass estimates. Due to current observational limitations, including instrumental constraints and the extreme distances separating Earth from these exoplanets and their host stars, astronomers are often unable to precisely measure or calculate the mass of many exoplanets nor determine their surface and atmospheric composition. This limits definitive habitability classification and heavily limits Earth-Like similarity assessments. In order to effectively determine Earth-Like planet composition, additional factors must be considered such as orbital eccentricity, stellar metallicity, stellar flux (radiation), total surface pressure, rotation period, axis inclination, ocean coverage, and atmosphere composition.

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